

Computer Architecture and Organization

Karnaugh Maps

(Important points on grouping of squares)

First, the concept of adjacency can be extended to include wrapping around the edge of the map. Thus, the top square of a column is adjacent to the bottom square, and the leftmost square of a row is adjacent to the rightmost square. These conditions are illustrated in Figures 1.1b and c.

Second, we can group not just 2 squares but 2^n adjacent squares (i.e., 2, 4, 8, etc.). The next three examples in Figure 1.1 show groupings of 4 squares. Note that in this case, two of the variables can be eliminated. The last three examples show groupings of 8 squares, which allow three variables to be eliminated.

We can summarize the rules for simplification as follows:

1. Among the marked squares (squares with a 1), find those that belong to a unique largest block of 1, 2, 4, or 8 and circle those blocks.
2. Select additional blocks of marked squares that are as large as possible and as few in number as possible, but include every marked square at least once. The results may not be unique in some cases. For example, if a marked square combines with exactly two other squares, and there is no fourth marked square to complete a larger group, then there is a choice to be made as to which of the two groupings to choose. When you are circling groups, you are allowed to use the same 1 value more than once.
3. Continue to draw loops around single marked squares, or pairs of adjacent marked squares, or groups of four, eight, and so on in such a way that every marked square belongs to at least one loop; then use as few of these blocks as possible to include all marked squares.

Figure 1.2a, is an example that illustrates the simplification process. If any isolated 1s remain after the groupings, then each of these is circled as a group of 1s.

Finally, before going from the map to a simplified Boolean expression, any group of 1s that is completely overlapped by other groups can be eliminated. This is shown in Figure 1.2b. In this case, the horizontal group is redundant and may be ignored in creating the Boolean expression.

One additional feature of Karnaugh maps needs to be mentioned. In some cases, certain combinations of values of variables never occur, and therefore the corresponding output never occurs. These are referred to as “don’t care” conditions. For each such condition, the letter “d” is entered into the

corresponding square of the map. In doing the grouping and simplification, each “d” can be treated as a 1 or 0, whichever leads to the simplest expression. An example is shown in Table 1.1, Truth Table for the One-Digit Packed Decimal Incrementer.

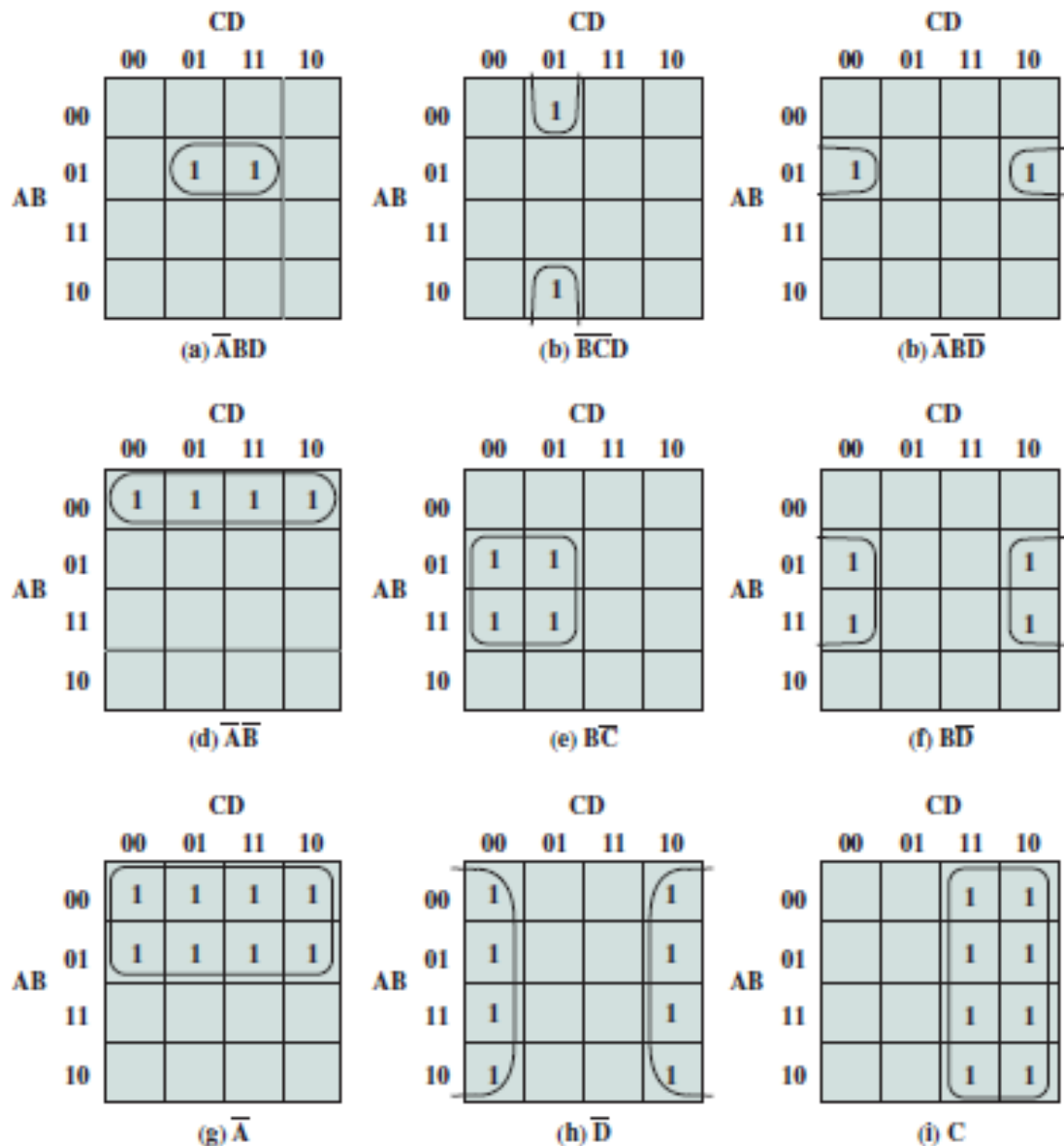


Figure 1.1: The Use of Karnaugh Maps

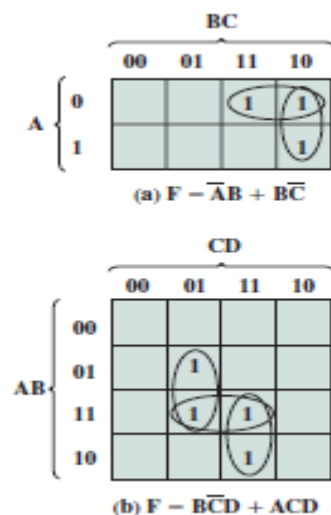


Figure 1.2: Overlapping Groups

Table 1.1: Truth Table for the One-Digit Packed Decimal Incrementer

Number	Input				Number	Output			
	A	B	C	D		W	X	Y	Z
0	0	0	0	0	1	0	0	0	1
1	0	0	0	1	2	0	0	1	0
2	0	0	1	0	3	0	0	1	1
3	0	0	1	1	4	0	1	0	0
4	0	1	0	0	5	0	1	0	1
5	0	1	0	1	6	0	1	1	0
6	0	1	1	0	7	0	1	1	1
7	0	1	1	1	8	1	0	0	0
8	1	0	0	0	9	1	0	0	1
9	1	0	0	1	0	0	0	0	0
Don't care condition	1	0	1	0		d	d	d	d
	1	0	1	1		d	d	d	d
	1	1	0	0		d	d	d	d
	1	1	0	1		d	d	d	d
	1	1	1	0		d	d	d	d
	1	1	1	1		d	d	d	D

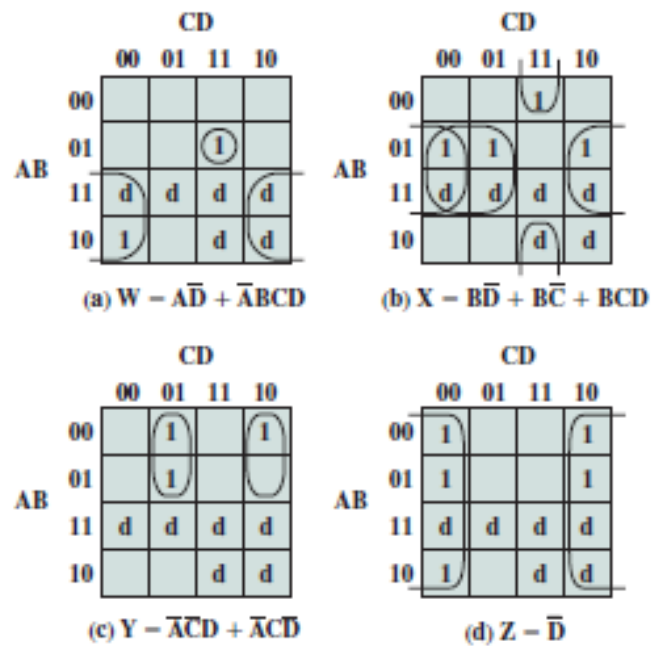


Figure 1.3: Karnaugh Maps for the Incrementer